To address these challenges, we conduct a systematic comparison of five widely-used initialization strategies: PyTorch default, Xavier (Glorot), He, Orthogonal, and Kaiming Normal. We evaluate these methods across four arithmetic operations in finite fields: modular addition, subtraction, division, and permutation composition. Our experimental setup employs a small Transformer architecture with 2 layers, 128 model dimensions, and 4 attention heads, allowing for controlled and reproducible investigations of grokking behavior.

Our main contributions are as follows:

- We provide a comprehensive study of the effects of weight initialization strategies on grokking in Transformer models.
- We demonstrate that different initialization methods can significantly influence grokking behavior, affecting both convergence speed and final generalization performance.
- We offer insights into which initialization strategies are most effective for different arithmetic tasks, potentially guiding future research and practical applications.
- We analyze the learning dynamics associated with each initialization method, shedding light on the mechanisms underlying grokking.

Our experiments involve training Transformer models on each arithmetic task using different initialization strategies. We carefully monitor training and validation performance, paying particular attention to sudden improvements in generalization that characterize grokking. Our results reveal that Xavier initialization often leads to faster convergence, particularly for tasks like modular addition and permutation composition. For instance, in the modular addition task (x_plus_y), Xavier initialization achieved 99% validation accuracy in just 863 steps, compared to 2363 steps for the baseline. Orthogonal initialization showed task-dependent performance, excelling in some operations but struggling in others.

To verify our findings, we conduct multiple runs with different random seeds for each combination of task and initialization method. We perform statistical analysis, including calculating 95% confidence intervals for key metrics such as steps to 99% validation accuracy. This approach ensures the robustness and reliability of our results.

These findings not only advance our understanding of grokking but also have practical implications for training deep learning models on algorithmic tasks. By optimizing weight initialization strategies, we may be able to induce grokking more reliably or accelerate the learning process. Our results suggest that the choice of initialization method can significantly impact both the speed of convergence and the final generalization performance, with some methods showing consistent advantages across multiple tasks.

Future work could explore several promising directions:

- Investigating the scalability of our findings to larger models and more complex tasks.
- Analyzing the interaction between initialization strategies and other hyperparameters, such as learning rate schedules or model architectures.
- Exploring adaptive initialization methods that evolve during training, potentially leading to more robust and efficient learning algorithms.
- Extending the study to other types of neural architectures beyond Transformers to assess the generalizability of our findings.

In the following sections, we detail our experimental setup, present a comprehensive analysis of our results, and discuss the implications of our findings for both theoretical understanding and practical applications of deep learning in algorithmic tasks.

2 Related Work

Our study intersects with several key areas of deep learning research: weight initialization strategies, the grokking phenomenon, and Transformer model training dynamics. This section compares and contrasts our approach with existing work in these domains.

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2.1 Weight Initialization Strategies

Weight initialization plays a crucial role in training deep neural networks, significantly impacting convergence speed and model performance. Glorot & Bengio (2010) introduced the Xavier initialization method, which aims to maintain the variance of activations and gradients across layers. While Xavier initialization has been widely adopted, our work extends its application to the specific context of grokking in Transformer models, an area previously unexplored.

He et al. (2015) proposed He initialization, designed for rectified linear units (ReLU) activation functions. Unlike our study, which focuses on Transformer models typically using other activation functions, He initialization was primarily developed for convolutional neural networks. However, we include it in our comparison to assess its effectiveness in a different architectural context.

Orthogonal initialization, proposed by Saxe et al. (2013), initializes weight matrices as random orthogonal matrices. While Saxe et al. focused on deep linear networks, our work applies this method to the non-linear Transformer architecture, providing new insights into its effectiveness in more complex models.

Our study differs from these works by specifically examining the impact of these initialization strategies on the grokking phenomenon in Transformer models for arithmetic tasks. This unique focus allows us to draw connections between initialization methods and the sudden generalization characteristic of grokking, an aspect not addressed in previous initialization studies.

2.2 Grokking Phenomenon

The grokking phenomenon, first described by Power et al. (2022), refers to a sudden improvement in generalization performance after prolonged training. While Power et al. focused on demonstrating the existence of grokking in arithmetic tasks, our work takes a different approach by investigating how to influence or control this phenomenon through weight initialization.

Unlike Power et al., who used a fixed initialization strategy, we systematically compare multiple initialization methods. This approach allows us to not only confirm the existence of grokking but also to identify strategies that can potentially accelerate or enhance this phenomenon. Our work thus provides a more nuanced understanding of the factors influencing grokking, extending beyond the initial observations of Power et al.

2.3 Transformer Training Dynamics

Transformer models (Vaswani et al., 2017) have become fundamental in many machine learning tasks. While Vaswani et al. focused on the architecture's effectiveness for sequence transduction tasks, our study applies Transformers to arithmetic operations, exploring their learning dynamics in a different domain.

Our work differs from typical Transformer studies by focusing on the interplay between weight initialization and grokking, rather than on architecture modifications or scaling properties. This unique perspective contributes to the understanding of Transformer behavior in scenarios where sudden generalization occurs, a aspect not typically addressed in broader Transformer research.

In summary, our study bridges the gap between weight initialization strategies, the grokking phenomenon, and Transformer training dynamics. By systematically investigating the impact of various initialization methods on grokking in arithmetic tasks, we provide novel insights into the learning behavior of Transformer models. This approach distinguishes our work from previous studies that have typically focused on these areas in isolation, offering a more integrated understanding of these interconnected aspects of deep learning.

3 Background

The Transformer architecture Vaswani et al. (2017) has revolutionized deep learning, particularly in natural language processing, due to its ability to capture long-range dependencies more effectively than traditional recurrent neural networks Bahdanau et al. (2014). Transformers use self-attention

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mechanisms to process input sequences, enabling parallel computation and improved performance on various tasks.

Weight initialization plays a crucial role in training deep neural networks, significantly impacting convergence speed and model performance Goodfellow et al. (2016). Several strategies have been proposed to address the challenges of training deep networks:

- Xavier (Glorot) initialization Glorot & Bengio (2010): Aims to maintain the variance of activations and gradients across layers.
- He initialization He et al. (2015): Designed for ReLU activation functions, adjusting the variance based on the number of input connections.
- Orthogonal initialization Saxe et al. (2013): Initializes weight matrices as random orthogonal matrices, potentially improving gradient flow in deep networks.
- Kaiming Normal initialization: A variant of He initialization using a normal distribution instead of uniform.

The grokking phenomenon, described by Power et al. (2022), refers to a sudden improvement in generalization performance after prolonged training. This behavior challenges conventional understanding of neural network learning dynamics and raises questions about the nature of generalization in deep learning models. Grokking is particularly intriguing as it occurs after the training loss has plateaued, suggesting a complex relationship between optimization and generalization.

3.1 Problem Setting

We consider a Transformer model f_θ with parameters θ, trained on a set of arithmetic tasks $\mathcal{T} = \{T_1, \ldots, T_n\}$. Each task T_i is defined as an operation over a finite field $\mathbb{F}_p$, where $p = 97$ (a prime number). The model receives input sequences $x = (x_1, \ldots, x_m)$, where each $x_j \in \mathbb{F}_p$, and is trained to predict the result of the arithmetic operation $y \in \mathbb{F}_p$.

We focus on four specific tasks:

- Modular addition (x_plus_y): $(a + b) \mod p$
- Modular subtraction (x_minus_y): $(a - b) \mod p$
- Modular division (x_div_y): $(a \cdot b^{-1}) \mod p$, where b^{-1} is the modular multiplicative inverse of b
- Permutation composition: Composition of two permutations of 5 elements

These tasks provide a controlled environment for studying neural network learning behavior, offering a clear distinction between memorization and true generalization.

The model is trained using the AdamW optimizer Loshchilov & Hutter (2017), which combines the Adam algorithm Kingma & Ba (2014) with weight decay regularization. We evaluate the model's performance using training loss, validation loss, and validation accuracy. Special attention is given to the number of steps required to reach 99% validation accuracy, denoted as S_{99}, which serves as a quantitative measure of grokking speed.

Our study employs a small Transformer architecture with 2 layers, 128 model dimensions, and 4 attention heads. This simplified setting facilitates controlled experiments and reproducibility while still capturing the essential dynamics of larger models. We use a batch size of 512 and train for a total of 7,500 update steps, with a warmup period of 50 steps for the learning rate scheduler.

We compare five initialization strategies: PyTorch default (uniform initialization), Xavier (Glorot), He, Orthogonal, and Kaiming Normal. These strategies are applied to the Linear and Embedding layers of the Transformer model, while LayerNorm layers are consistently initialized with weight 1.0 and bias 0.0 across all experiments.

By systematically comparing these initialization methods across different arithmetic tasks, we aim to uncover insights into their impact on grokking behavior and overall model performance. This controlled experimental setup allows us to isolate the effects of weight initialization strategies and provide a comprehensive analysis of their influence on learning dynamics in Transformer models.

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